RAJALAKSHMI ENGINEERING COLLEGE

Vathisha R 251801369-2025-AIDS

V2

Started on	Monday, 24 August 2026, 1:03 PM
State	Finished
Completed on	Monday, 24 August 2026, 1:17 PM
Time taken	13 mins 25 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)
{
    int i= 1;
```

```
    int s =1;
```

```
    while(s <= n)
    {
         i++;
         s += i;
     }
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

For example:

Input	Result
9	12

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
int main(){
    int n;
    scanf("%d",&n);
    int i=1;
    int s=1;
    int count=2;
    while(s<n){
        count++;
        i++;
        count++;
        s=s+i;
        count++;
    }
    count++;
    printf("%d",count);
    return 0;
}
```

	Input	Expected	Got	
	9	12	12	
	4	9	9	

Passed all tests!

Correct

Marks for this submission: 1.00/1.00.

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V2

Started on	Monday, 24 August 2026, 1:17 PM
State	Finished
Completed on	Monday, 24 August 2026, 1:24 PM
Time taken	6 mins 35 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void func(int n)
{
    if(n==1)
    {
      printf("*");
    }
    else
    {
     for(int i=1; i<=n; i++)
     {
       for(int j=1; j<=n; j++)
       {
          printf("*");
          printf("*");
          break;
       }
     }
   }
 }
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer: (penalty regime: 0 %)

```c
#include<stdio.h>
int main(){
    int n;
    scanf("%d",&n);
    int count;
    if(n==1)
        count=2;
    else
        count=5*n+2;
    printf("%d",count);
    return 0;
}
```

	Input	Expected	Got	
	2	12	12	
	1000	5002	5002	
	143	717	717	

Passed all tests!

Correct

Marks for this submission: 1.00/1.00.

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V2

Started on	Monday, 24 August 2026, 1:31 PM
State	Finished
Completed on	Monday, 24 August 2026, 1:37 PM
Time taken	6 mins 10 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {
{
   for (i = 1; i <= num;++i)
   {
    if (num % i== 0)
       {
         printf("%d ", i);
       }
    }
 }
```

Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```c
#include<stdio.h>
int main(){
    int num;
    scanf("%d",&num);
    int count=1;
    for(int i=1;i<=num;i++){
        count++;
        if(num%i==0){
            count++;
        }
        count++;
    }
    printf("%d",count);
    return 0;
}
```

	Input	Expected	Got	
☐	12	31	31	☐
☐	25	54	54	☐
☐	4	12	12	☐

Passed all tests!

Correct

Marks for this submission: 1.00/1.00.

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V2

Started on	Monday, 24 August 2026, 1:51 PM
State	Finished
Completed on	Monday, 24 August 2026, 3:46 PM
Time taken	1 hour 54 mins
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time

complexity using counter method.

```
void function(int n)
{
    int c= 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```c
#include<stdio.h>
int main(){
    int n;
    scanf("%d",&n);
    int count=1;
    for(int i=n/2;(++count) && i<n;i++){
        for(int j=1;(++count) && j<n;j=2*j){
            for(int k=1;(++count)&& k<n;k=2*k){
                count++;
            }
        }
    }
    printf("%d",count);
    return 0;
}
```

	Input	Expected	Got	
☐	4	30	30	☐
☐	10	212	212	☐

Passed all tests! ☐

Correct

Marks for this submission: 1.00/1.00.

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V2

Started on	Monday, 24 August 2026, 3:46 PM
State	Finished
Completed on	Monday, 24 August 2026, 3:52 PM
Time taken	5 mins 9 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
void reverse(int n)
{
   int rev = 0, remainder;
   while (n != 0)
    {
        remainder = n % 10;
        rev = rev * 10 + remainder;
        n/= 10;

    }
print(rev);
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```c
#include<stdio.h>
int main(){
    int n;
    scanf("%d",&n);
    int rev=0,remainder;
    int count=1;
    while(1){
        count++;
        if(n<=0)
            break;
        remainder=n%10;
        count++;
        rev=rev*10+remainder;
        count++;
        n=n/10;
        count++;
    }
    count++;
    printf("%d",count);
    return 0;
}
```

	Input	Expected	Got	
☐	12	11	11	☐
☐	1234	19	19	☐

Passed all tests!

Correct

Marks for this submission: 1.00/1.00.

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